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Compressive and Tensile Mechanical Response of a 316L Stainless Steel at Low, Intermediate, and High Strain Rates

Angela Y. Ku¹, Pooyan Javadzadeh^{1,2}, Jie Sheng², Xu Nie², Wei Tong², Bo Song^{1*}

¹Sandia National Laboratories, 1515 Eubank SE, Albuquerque, NM, 87123, USA

²Southern Methodist University, Department of Mechanical Engineering, Dallas, TX, 75206, USA

*Corresponding author: bsong@sandia.gov

Abstract

A comprehensive study of the mechanical response of 316 stainless steel in compression and tension at room temperature and at various strain rates is presented. A universal testing machine and a split-Hopkinson bar technique were used to determine the mechanical behavior at quasi-static and dynamic strain rates under tension and compression loading, while a Drop-Hopkinson bar was used to characterize the tensile behavior at an intermediate strain rate of 170 s^{-1} . The effects of strain rate on flow stress, true tensile failure strain and onset of necking were determined for the 316L stainless steel. Both modified Johnson-Cook (JC) models (JC-2019 and JC-2026) described the strain-rate-dependent stress-strain response of 316L stainless steel very well. The JC-2026 model was also used to inversely calculate strain-rate and strain-dependent Taylor-Quinney (TQ) coefficients as well as specific stored energy for the 316L stainless steel. The specific stored energy showed an uptrend with increasing strain rate and plastic strain, indicating more energy contributes to microstructural change rather than generating heat.

Keywords: *Stainless steel, strain rate, Kolsky bar, Johnson-Cook model, Taylor-Quinney coefficient, stored energy*

1 Introduction

Stainless steels have been of great interest to automotive, aerospace, defense, medical, and nuclear power industries for their excellent corrosion resistance, machinability, and mechanical performance [1] [2] [3]. In these applications, it is desirable for stainless steels to be resistant against external impact ranging from low-speed accidental crash to explosion. The resistance performance requires understanding of the mechanical response under different mechanical loading environments. The 300 series stainless steels are non-magnetic when annealed and can only be strengthened through cold working. They have a face centered cubic (FCC) austenitic structure that is attained through alloying nickel, manganese, and nitrogen [4], and transforms to a body centered cubic (BCC) martensitic structure during mechanical loading [5] [6] [7] [8] [9] [10] [11]. The type and amount of deformation-induced martensite depend on the chemical composition, grain size, stacking fault energy, deformation temperature, strain rate, and stress state. Among the 300 series stainless steels, 304 is the most common and cost-effective grade. However, with the addition of molybdenum, 316 is the superior choice against corrosion for harsh, salty, or acidic environments.

The mechanical responses of austenitic stainless steels under the conditions of tension [12] [13] [14] [15] and compression [16] [17] [18] loading have been investigated by several research groups over the years. Compared to extensively characterized 304 stainless steel, the mechanical properties of 316 stainless steel are relatively scarce. Most of the mechanical characterization was conducted at quasi-static strain rates. As early as the 1970s, Steichen was one of the scholars looking into the effect of strain rate on mechanical properties of 316 stainless steel [19]. He concluded that 316 stainless steel “can be very rate sensitive over certain temperature and strain-rate regimes” [19]. Kolsky bar, a.k.a. split Hopkinson bar, has been recognized as an effective tool for high-strain-rate mechanical characterization in compression, tension, or shear [20], making it possible to investigate strain-rate effects from the quasi-static to dynamic regimes. The Kolsky bar has recently been used to characterize compressive response of 316 stainless steels at high strain rates up to 6000 s^{-1} [21] [22] [23] [24] [25]. However, very limited dynamic tensile characterization has been conducted for 316 stainless steel. Jia et al. [18] employed a Kolsky tensile bar to characterize the tensile stress-strain response of 316 stainless steel at various strain rates for determining material constants in the Johnson-Cook (JC) model. Albertini and his colleagues [26] [27] conducted dynamic shear, uniaxial and biaxial tensile tests on 316 stainless steel for understanding the effect of stress state on the material’s response. Nonetheless, there has been no such work published for a full spectrum mechanical characterization of 316 stainless steel from low, intermediate, to high strain rates.

For constitutive characterization involving both tension and compression, it is particularly important to consider whether 316L exhibits loading-mode asymmetry. El-Tahawy et al. [10] directly compared room-temperature tensile and compressive responses of 316L stainless steel and found that the tensile flow stress exceeded the compressive flow stress by only approximately 40

MPa up to a true strain of approximately 0.25; however, between true strains of 0.25 and 0.50, the tensile material hardened substantially more rapidly, producing a tensile flow stress approximately 200 MPa higher at a true strain of 0.50. Their microstructural analysis attributed this asymmetry to differences in texture evolution, dislocation density, and martensitic phase fraction. The tension–compression difference is physically plausible because transformation from austenite to α' -martensite involves volumetric expansion and may therefore be promoted differently under tensile and compressive stress states. Characterizing the material in compression and tension at similar strain rates also helps understand the symmetry of material response in compression and tension at different strain rates, which provides valuable insight to strain-rate-dependent material model development [28].

The complexity of the observed material response motivates the development of constitutive models suitable for numerical simulation. A number of constitutive models have been proposed over the years to describe the mechanical behavior of austenitic stainless steels [16] [18] [29] [30] [31]. They can be divided into two categories: empirical phenomenological models such as the JC model [32], and microstructure based physical models such as the Zerilli–Armstrong model [33]. Each type of model is unique in its complexity and accuracy in capturing the available set of experimental data. A summary of all the phenomenological and physical models that have been developed, along with their advantages and disadvantages, can be found in the review article by Lin and Chen [34]. The JC model has remained one of the most widely implemented phenomenological constitutive laws because of its compact mathematical form, accessible calibration procedure, and availability in commercial finite element codes. Gupta et al. [35] compared JC, modified Zerilli–Armstrong, modified Arrhenius, and artificial-neural-network descriptions for austenitic stainless steel 316 at temperatures from 323 to 623 K and strain rates from 10^{-4} to 10^{-1} s⁻¹. Their results showed that the JC model provided a lower correlation coefficient than the modified Zerilli–Armstrong and modified Arrhenius forms, indicating limited predictive capability for the studied thermomechanical range. Malakizadi et al. [36] reported that the classical JC model overestimated thermal softening for 316L stainless steel within portions of the high-temperature deformation range and consequently underestimated flow stress, heat generation, and cutting forces. Seo et al. [31] modified the JC relation for 316L stainless steel to incorporate strain–rate/strain interaction and reduced the error relative to the classical formulation. These results directly support the use of modified constitutive treatments for 316L rather than relying exclusively on the original JC form.

As presented in [37], the major modifications to the classic JC model include the effects of strain rate and thermosoftening due to adiabatic heating at high strain rates. Song and Sanborn [37] modified the original JC model with a power law form for strain rate effect and an explicit form of strain-rate-dependent adiabatic thermosoftening regardless of strain. However, the explicit thermosoftening form was fully empirical without direct experimental data support. Recently, Song et al. [38] developed a semi-empirical form of strain-rate- and strain-dependent Taylor-

Quinney (TQ) coefficient, which was then implemented into a further modified JC model [39]. This implementation provides a great opportunity to describe the strain-rate-dependent stress-strain response of 316 stainless steel with the most recently modified JC model.

The objective of the current work is to characterize the tension and compression behavior of 316L stainless steel, which possesses a lower carbon content for better weldability in comparison with 304 and 316 stainless steels, at low, intermediate, and high strain rates using a universal testing machine, Drop-Hopkinson bar, and Kolsky bar technique, respectively. The effect of strain rate on the mechanical response of 316L stainless steel is determined. The compressive and tensile stress-strain responses were compared at similar strain rates for investigating the compression/tension symmetry for 316L stainless steel. The experimental data were used to fit into the modified Johnson-Cook model (JC-2019) by Song and Sanborn [37] and the further modified version (JC-2026) that was developed in 2026 [39]. Based on the most recent version of the JC-2026 model, the strain and strain-rate-dependent TQ coefficient was determined for the 316L stainless steel. The experimental data and model correlations obtained from this study provide a solid foundation for further analysis of 316L stainless steel on the component and system design levels for various applications in automotive, aerospace, defense, medical, and nuclear power industries.

2 Material and Experiments

2.1 Material

The annealed 316L stainless steel under this investigation was obtained from Marco Steel & Aluminum. The chemical composition from the supplier is listed in Table 1. The tension and compression specimens were fabricated from a cold-drawn round bar with a diameter of 25.4 mm. The tension specimens had a diameter of 3.18 mm and gage length of 6.35 mm, for an L/D ratio of 2, as shown in Fig. Tension specimen

(a). The compression specimens had a diameter of 6.35 mm and length of 3.18 mm for an L/D ratio of 0.5, as shown in Fig. 1(b). The specimens were designed to ensure uniform deformation and to reduce the longitudinal and radial inertia effects during a dynamic experiment.

Table 1. Reported chemical composition of 316 stainless steel

C	Co	Cr	Cu	Mn	Mo	N	Ni	P	S	Si	Fe
0.015	0.41	16.53	0.36	1.29	2.024	0.030	10.51	0.031	0.026	0.27	Remainder

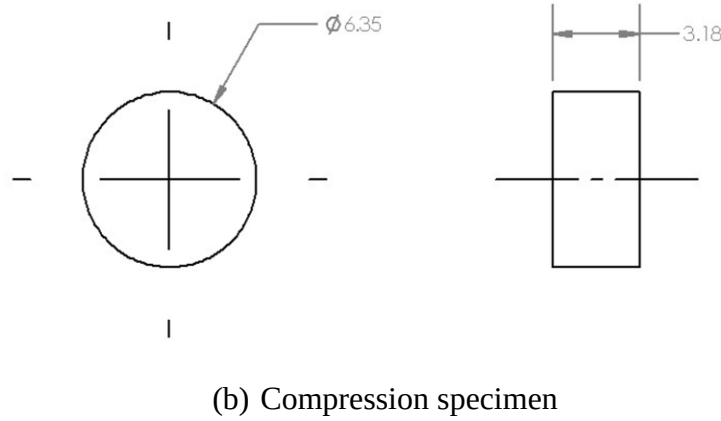
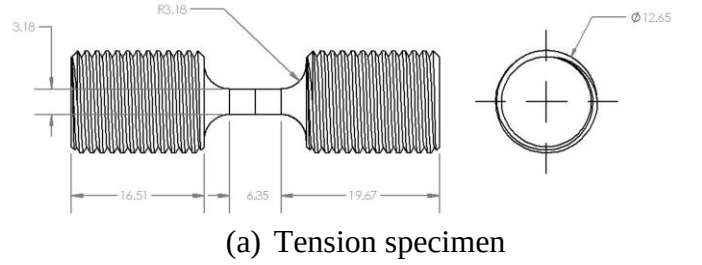


Figure 1. 316L stainless steel specimen designs. (a) tension; (b) compression. Dimensions are in mm.

2.2 Uniaxial quasi-static tension and compression experiments

Quasi-static tension and compression experiments were performed at Southern Methodist University using a 5967 Instron load frame with a 30-kN load cell and a 5582 Instron load frame with a 100-kN load cell, respectively, at two nominal strain rates of 10^{-3} s^{-1} and 10^{-1} s^{-1} .

For the tension experiments, the samples were threaded into custom-made cylindrical stainless-steel grips with a diameter of 38.1 mm and a length of 69.85 mm. The grips were used to connect the samples to the crosshead and base beam adapters of the 5967 Instron load frame as shown in Fig. 2. The crosshead moved at a constant speed of either 0.00635 mm/s or 0.65 mm/s to pull the tension specimen to failure and to achieve a nominal strain rate of 10^{-3} s^{-1} and 10^{-1} s^{-1} , respectively. A video extensometer program based on digital image correlation developed by Tong et al. [40] [41] [42] was used to measure the in-situ full-field deformation. As shown in Fig. 2 (right), prior to the testing the surface of each sample was sprayed within its gage section with fine black paint speckles over a thin coat of white paint. During tensile deformation several hundred 8-bit grayscale digital images (1600-by-500 pixels) were acquired at a rate of either 1 frame per second or 50 frames per second, depending on the strain rate. The typical pixel resolution for the digital images

was ~9 microns/pixel. The true tensile strain was determined over a 4 mm portion at the center of the gage section of each tensile specimen. The axial true tensile stress was obtained from the axial load corresponding to each image divided by the real-time cross-section area of the tensile specimen per the usual assumption of plastic incompressibility.

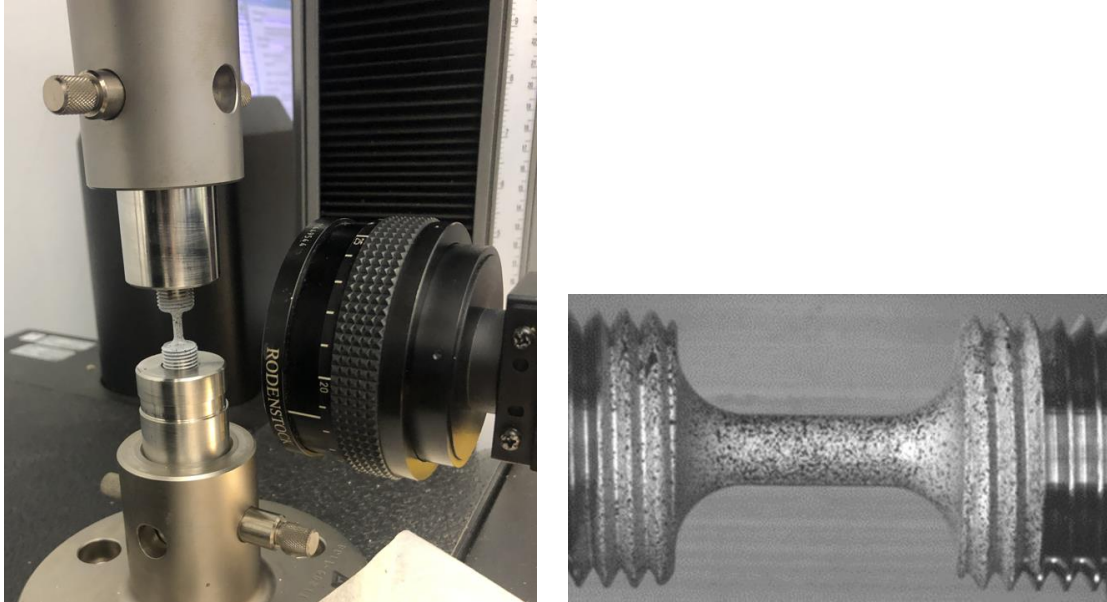


Figure 2. A typical experimental setup for the tensile test (left): a 316 stainless steel tension specimen with sprayed black paint speckles (right) is threaded into the tension grips and a CCD camera with a lens is placed at the side for acquiring digital images.

For the compression experiments, the samples were sandwiched as shown in Fig. 3 between two tungsten carbide (WC) platens. The WC platens had a diameter of 25.4 mm and a thickness of 4.06 mm and were mounted to 38.1-mm diameter stainless steel loading rods that were linked to either the crosshead adapter or the base beam adapter of the 5582 Instron load frame. A thin Teflon film was placed on the top and bottom platens to minimize the interfacial friction of the specimen. A crosshead speed of either 0.0048 mm/s or 0.48 mm/s was imposed to obtain a nominal strain rate of 10^{-3} s^{-1} and 10^{-1} s^{-1} , respectively. The axial deformation of the compressed specimen was obtained by removing the system compliance from the total crosshead displacement. The computed axial deformation was verified by comparing its final value against the measured length of the compression specimen after testing. Similar to the data analysis for the tension tests, the axial true compressive stress was obtained based on the same assumption of plastic incompressibility.



Figure 3. A 316 stainless steel compression specimen placed between two loading rods with WC platens prior to testing.

2.3 Uniaxial dynamic tension and compression experiments

Kolsky bars were used to perform dynamic uniaxial tension and compression experiments at nominal strain rates of 500 s^{-1} , 1500 s^{-1} , and 3000 s^{-1} . Schematics of the tension and compression setups are shown in Fig. 4 and presented in detail in [43] and [44], respectively. Both the tension and compression setups used in this study consist of a striker, incident bar, transmission bar, and momentum bar with diameters of 25.4 mm made of C300 maraging steel. The incident bar was 3.658 m long while the transmission bar was 2.134 m long. A variety of striker bar lengths and impact speeds, as well as pulse shapers, were used to achieve the desired strain and strain rate levels.

The tension setup consists of a solid striker located inside a tubular gun barrel (Fig. 4(a)). When compressed gas is released from the air cylinder, the striker impacts the end cap, which is threaded into the opposite end of the gun barrel, to generate a dynamic tensile pulse. A small piece of annealed copper, which serves as a pulse shaper, is placed on the end cap to remove the high-frequency oscillations of the tensile pulse and also to tune the profile of the incident pulse to ensure stress equilibrium and constant strain rate deformation in the specimen. The tensile stress pulse travels in the opposite direction of the striker into the incident bar, which is connected to the gun barrel through a coupler. When the tensile pulse reaches the tension specimen, part of the pulse is reflected into the incident bar while the rest is transferred into the transmission bar through the specimen. The tensile dog-bone specimen is threaded into the incident and transmission bar ends

and secured with lock nuts with a torque of 14.12 N·m to ensure a tight contact. Strain gages were mounted onto the incident and transmission bars at 1.83 m and 0.545 m from the specimen, respectively, to record the stress pulses in each bar. A high-speed laser extensometer, as shown in Fig. 4(a) and presented in detail in [45], was used to improve the accuracy of the strain measurements due to the threaded joint between the tension specimen and the bar ends. The laser beam was split into two separate high-speed photodetectors to measure the displacement histories of the incident and transmission bar ends individually. Since the laser extensometer measured the total displacement of the tension specimen, which included both the gage and non-gage sections, the deformations over the gage section for all tension specimens were corrected using the procedures presented in [46].

The compression setup is similar to the tension setup, with the dynamic compressive pulse generated by the direct impact of the striker on the end of the incident bar (Fig. 4(b)). A dual pulse-shaping technique was used, which consisted of a piece of annealed copper and pre-hardened steel disk placed on the impact end of the incident bar [47]. Upon impact, the annealed copper eliminates the high-frequency oscillations, while the steel disk tailors the overall shape of the incident pulse with increased work hardening so that constant strain rate can be achieved under dynamic stress equilibrium. The compressive pulse travels along the incident bar, and when it reaches the compression specimen, part of the pulse is reflected into the incident bar while the rest is transferred into the transmission bar. Strain gages were mounted onto the incident and transmission bars at 1.92 m and 0.459 m from the specimen, respectively, to record the stress pulses of each bar. A thin layer of Dow Corning high-vacuum grease was placed on the incident and transmission bar ends to minimize the interfacial friction between the compression specimen and the bar ends.

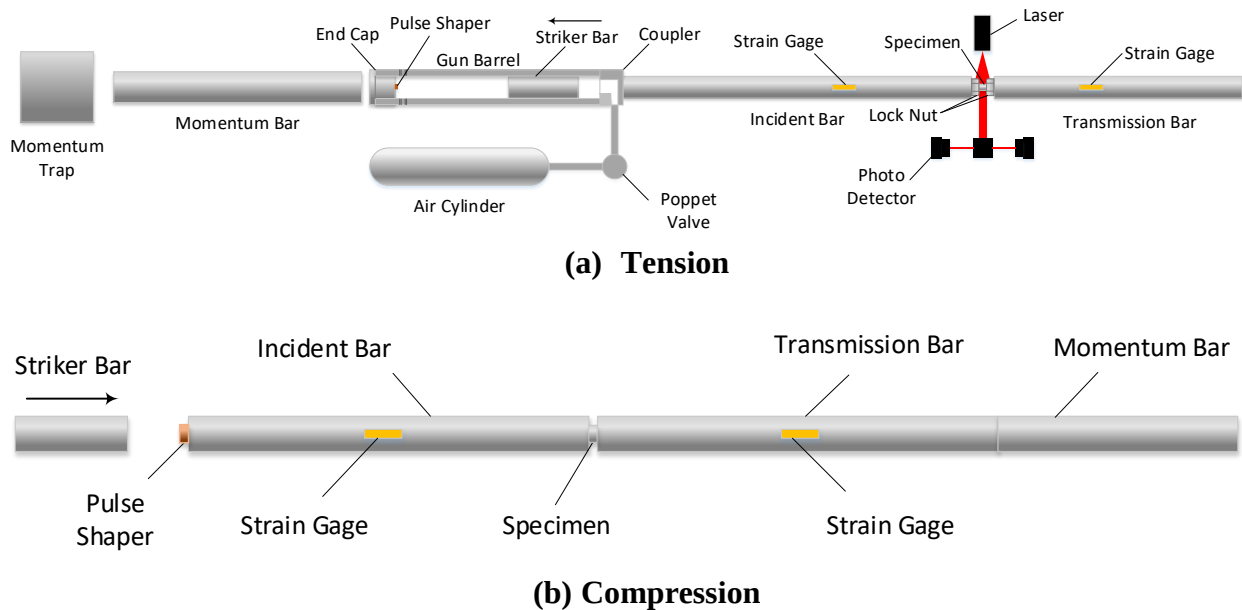


Figure 4. Schematic view of the Kolsky bar setup for (a) tension and (b) compression experiments.

Uniaxial tensile experiments at an intermediate strain rate of $\sim 200 \text{ s}^{-1}$ were conducted using a Drop-Hopkinson bar, as shown in Fig. 5. The Drop-Hopkinson bar is a combination of a drop table and Hopkinson bar, and the detailed procedures for the apparatus can be found in [48]. The specimen was threaded into an impact plate and a 3.658 m long transmission bar made of maraging C300 steel and was secured with lock nuts with a torque of 14.12 N·m. Low-speed impact is generated by dropping a carriage that is attached with an impactor onto an impact plate via a load spreader. A piece of rubber with a durometer of 75D was used as a pulse shaper that was placed on top of the load spreader to generate the desired profile for the incident pulse. The same measuring techniques and post-processing methods used for the conventional Kolsky tension bar experiments were adapted here to measure the displacement histories of the bars and to correct the specimen deformation over the gage section.

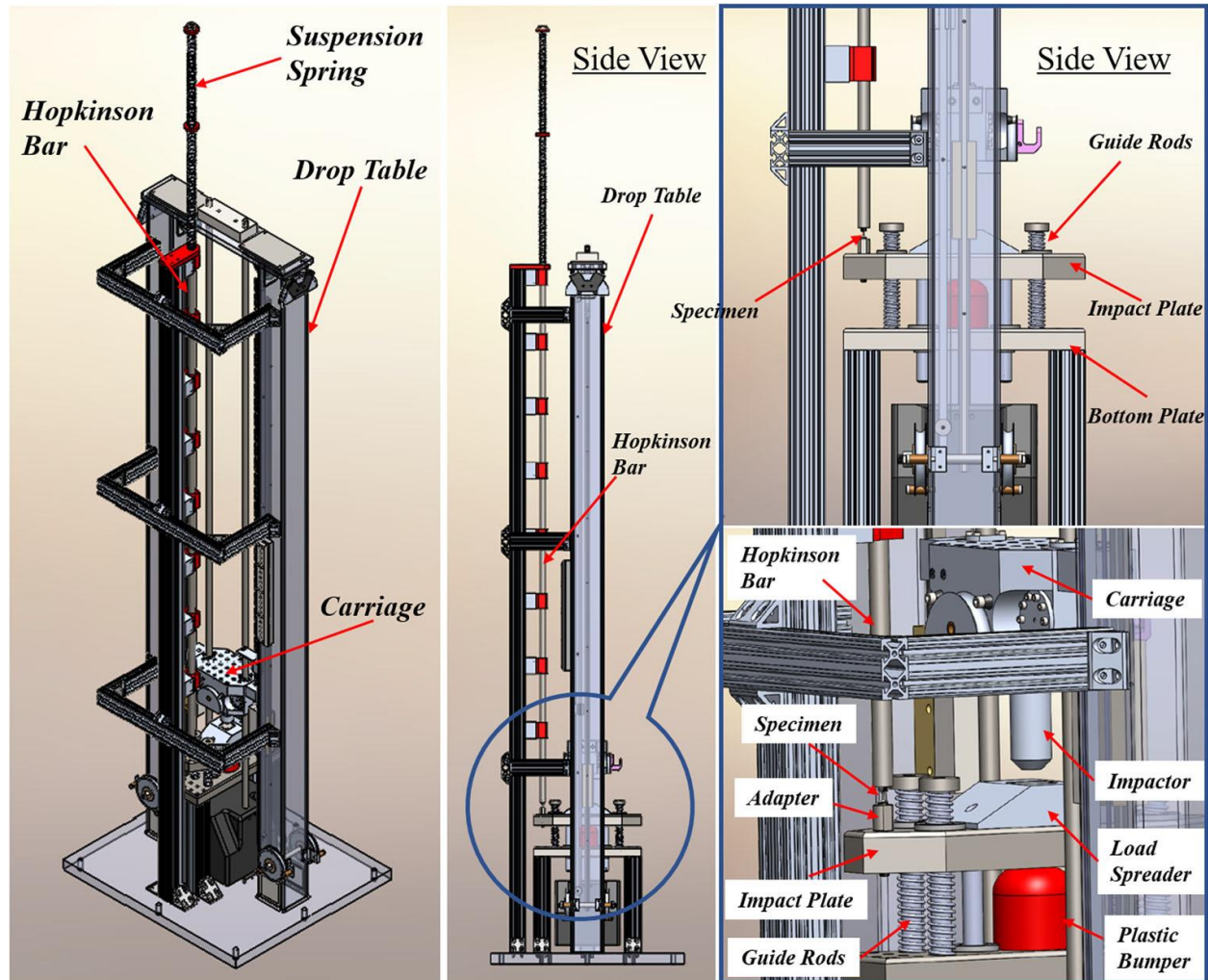


Figure 5. Schematic view of the Drop-Hopkinson bar setup.

3 Results and Discussion

The mechanical behavior of the 316L stainless steel was characterized in compression and tension at quasi-static, intermediate, and dynamic strain rates. At least three specimens were tested at each experimental condition, and the average engineering and true stress-strain curves are presented in Figs. 6 and 7, respectively.

No specimen failure/fracture was observed in compression tests. The engineering stress-strain curves in tension are plotted to failure. However, it should be noted that, due to the limited duration of loading, the stress-strain curve at 520 s⁻¹ as shown in Fig. 6 (a) does not represent specimen failure. The true tensile stress-strain curves at 10⁻³ s⁻¹ and 10⁻¹ s⁻¹ strain rates were calculated using a video extensometer program based on digital image correlation as mentioned in Section 2.2. The true stress-strain curves at 170 s⁻¹, 520 s⁻¹, 1430 s⁻¹, and 3190 s⁻¹ strain rates were calculated from the engineering stress-strain curves and are presented in equations (1) and (2) as the following:

$$\varepsilon_t = \ln(1 + \varepsilon_e) \quad (1)$$

$$\sigma_t = \sigma_e(1 + \varepsilon_e) \quad (2)$$

where ε_e and σ_e are the engineering strain and stress, respectively, and ε_t and σ_t are the true strain and stress, respectively. In Eqs. (1) and (2), both stress and strain take positive values in tension and negative values in compression. However, the stress and strain were plotted positive for both compression and tension and are clearly marked with annotations for comparison purposes. The true stress-strain curves in tension, as shown in Fig. 7(a), are plotted up to the onset of necking because Eqs. (1) and (2) are no longer valid after the specimen necks due to highly localized plastic deformation over the necking region.

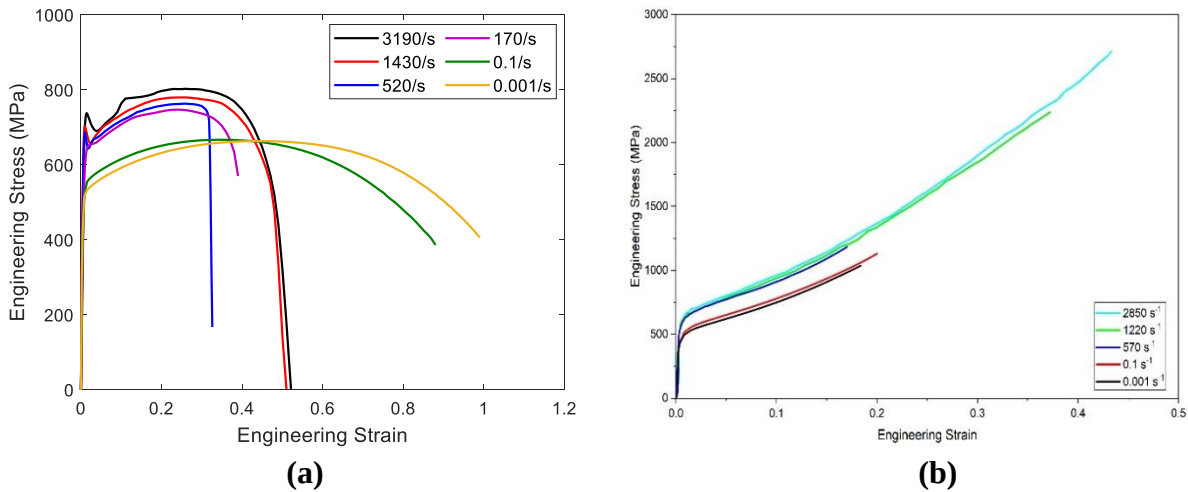


Figure 6. Engineering stress-strain behavior of 316L stainless steel at various strain rates in (a) tension and (b) compression.

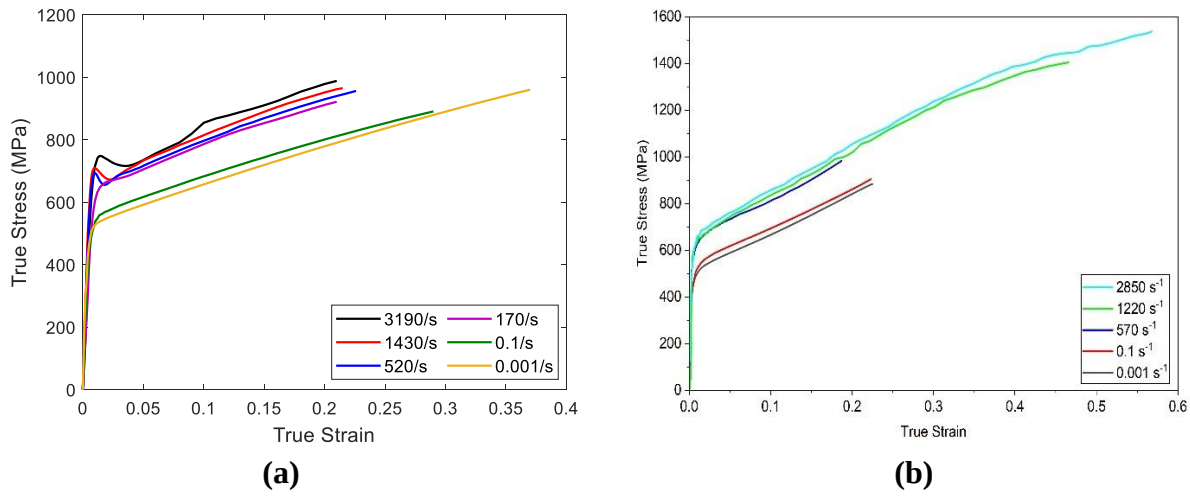


Figure 7. True stress-strain behavior of 316 stainless steel at various strain rates in (a) tension and (b) compression.

The strain rate effect on the 0.2% yield stress, as well as the flow stress at 10% and 20% true strain, are shown in Fig.8. The average of the upper and lower yield was used for the 0.2% yield stress in tension. The flow stress exhibits positive strain rate sensitivity that is most significant at strain rates above 10^{-1} s^{-1} . The amount of positive strain rate sensitivity also varies with strain. Increasing strain rate from 10^{-3} s^{-1} to 3000 s^{-1} results in an average increase of $\sim 45\%$ for the 0.2% yield stress and an average increase of $\sim 17\%$ for the flow stress at 20% strain. Overall, the strain rate sensitivity is slightly more prominent in tension than in compression.

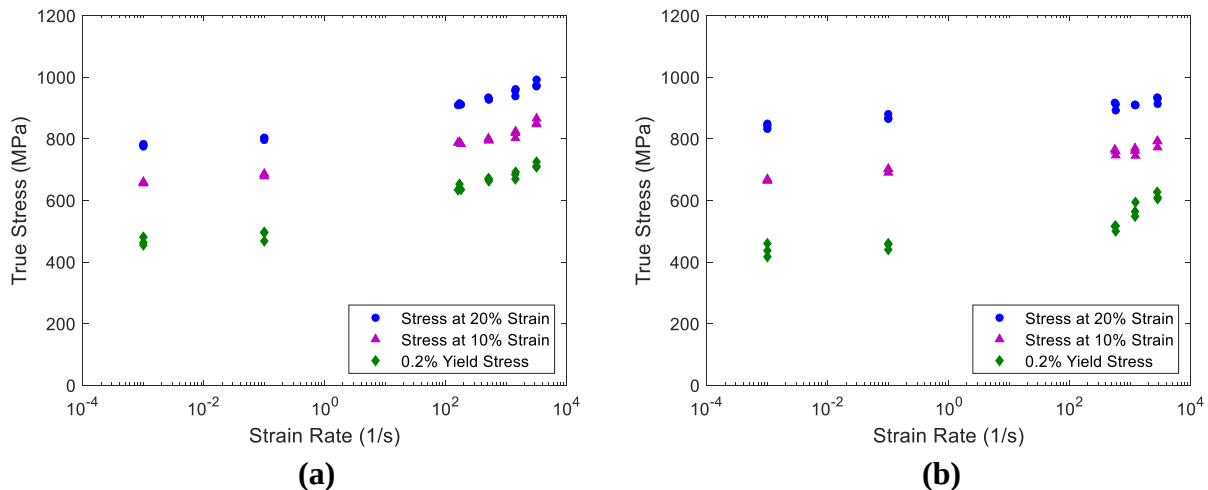


Figure 8. Strain rate effect of 316 stainless steel on flow stress at various strains in (a) tension and (b) compression.

The true tensile failure strains were determined by measuring the minimum diameter of the post-test tension specimens using a calibrated optical microscope and calculated using the following equation:

$$\varepsilon_f = \ln \frac{A_0}{A_f} \quad (3)$$

where ε_f is the true failure strain, A_0 is the initial cross-sectional area, and A_f is the minimal cross-sectional area at failure. A typical micrograph of the measurement is shown in Fig. 9. The strains at onset of necking were determined from the engineering strain at the ultimate tensile stress, which is then converted to true strain using Eq. (1).

Figure 10 shows the effect of strain rate on the true failure strains and true strains at the onset of necking from the tensile tests. Overall, the true failure strains and strains at the onset of necking exhibit strain rate dependence. For the 316L stainless steel, increasing strain rate from 10^{-3} s^{-1} to 3000 s^{-1} results in a $\sim 40\%$ decrease in true failure strain. The strains at the onset of necking decrease by $\sim 40\%$ when increasing strain rate from 10^{-3} s^{-1} to 200 s^{-1} but remain somewhat constant in the intermediate and dynamic strain rate range.

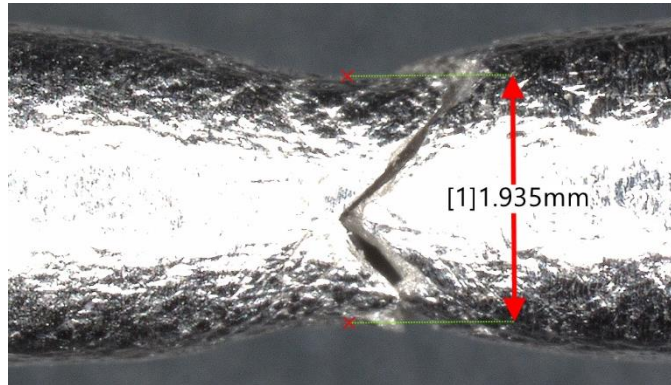


Figure 9. Optical micrograph of the minimum diameter measurement of a tension specimen at strain rate of 3000 s^{-1} using a calibrated optical microscope.

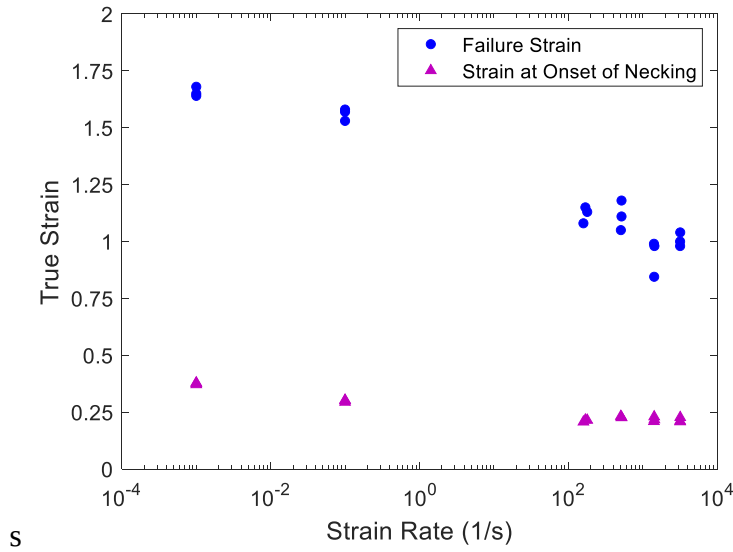


Figure 10. Strain rate effect of 316 stainless steel for true strain at the onset of necking and true failure strain.

Figure 11 compares the true tensile and compressive response under quasi-static and dynamic loading at similar strain rates. The dynamic response between 500 s^{-1} and 3000 s^{-1} strain rate shows reasonable symmetry in flow stress. A higher strain hardening rate is observed for the compression tests, and the tension-compression asymmetry becomes more apparent after $\sim 10\%$ strain. This difference is possibly due to the effect of interfacial friction between the specimen and bar ends in compression tests.

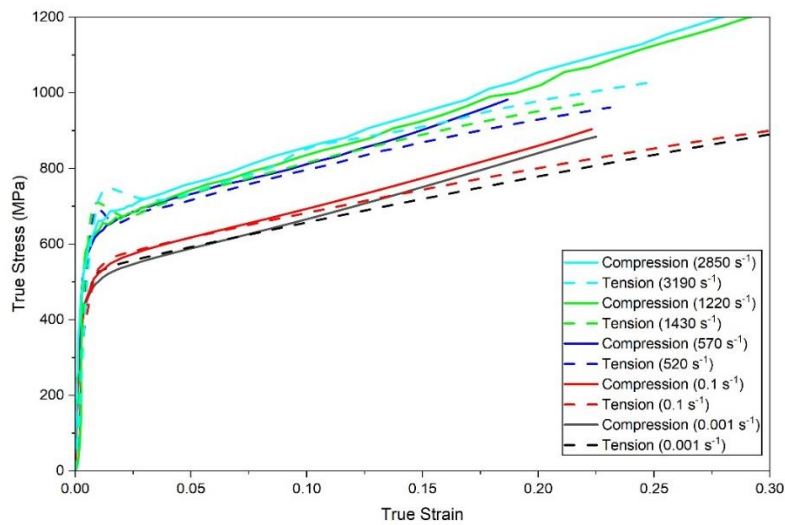


Figure 11. Comparison of the true tension and compression behavior of the 316L stainless steel.

4 Constitutive Modelling

Based on the obtained experimental data, the two different forms of JC models modified in 2019 [37] and 2026 [39], respectively, were used to describe the stress-strain response of the 316L stainless steel at various strain rates.

Both modified JC models were based on the same framework originally developed by Johnson and Cook [32],

$$\sigma = (A + B\varepsilon^n) \cdot g(\dot{\varepsilon}) \cdot h(T) = (A + B\varepsilon^n) \cdot (1 + C \ln \dot{\varepsilon}) \cdot \left(1 - \left(\frac{T-T_0}{T_m-T_0}\right)^m\right) \quad (4)$$

where σ , ε_p , $\dot{\varepsilon}$, and T are flow stress, plastic strain, strain rate, and temperature, respectively; A , B , C , m , and n are material constants; T_m and T_0 are melting and reference (usually ambient) temperatures, respectively. Both stress and strain take true measurements, whereas the strain rate usually refers to engineering strain rate [49]. In 2019, Song and Sanborn [37] modified the JC model with a power-law strain rate effect term as presented in Eq. (5),

$$g(\dot{\varepsilon}) = \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)^\alpha \quad (5)$$

where $\dot{\varepsilon}_0$ is reference strain rate. In order to account for thermosoftening effects due to adiabatic heating particularly at high strain rates, the temperature sensitivity term was replaced with an explicit function of strain and strain rate as the following:

$$h(T) = 1 - \left(\frac{D \cdot \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)^\alpha \cdot Y(\varepsilon)}{1 + \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{t0}}\right)^{-\frac{1}{\delta}}}\right)^m \quad (6)$$

where D and δ are constants; $\dot{\varepsilon}_{t0}$ is another reference strain rate; and

$$Y(\varepsilon) = \left(A + \frac{B}{n+1} \cdot \varepsilon^n\right) \cdot \varepsilon \quad (7)$$

The modified JC model (JC-2019) was thus presented as

$$\sigma = (A + B\varepsilon^n) \cdot \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)^\alpha \cdot \left(1 - \left(\frac{D \cdot \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)^\alpha \cdot Y(\varepsilon)}{1 + \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{t0}}\right)^{-\frac{1}{\delta}}}\right)^m\right) \quad (8)$$

It is noted that the JC-2019 model (Equation (8)) was based on an empirical strain-rate-dependent TQ regardless of strain. Recently, a semi-empirical strain-rate and strain dependent TQ has been developed [39], which was then used to further modify the JC model (JC-2026) [38] with the following form,

$$\sigma = (A + B\varepsilon^n) \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^\alpha \left\{ 1 - \left[C_0'' \left(1 - \frac{\dot{\varepsilon}_{t0}}{\dot{\varepsilon} + \dot{\varepsilon}_{t0}} \right) \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^{\alpha'} \varepsilon \right]^m \right\} \quad (9)$$

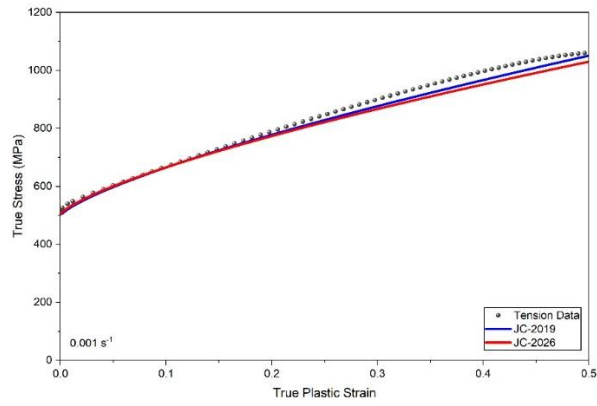
where α' and C_0'' are material constants. Compared to the JC-2019 and JC-2026 models, the JC-2026 model accounts for both strain rate and strain dependent TQ coefficient but with a simpler form.

In this study, the true tensile stress-strain curves up to the onset of necking, as presented in Fig. 7(a), were used to fit into the JC-2019 and JC-2026 models (Equations (8) and (9)). The material constants for both models are summarized in Table 2.

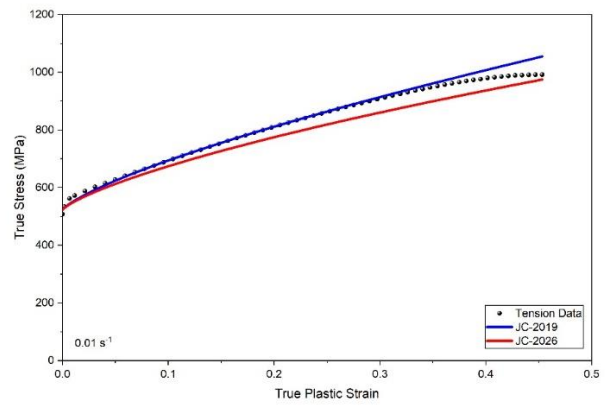
Table 2. Material Constants for JC-2019 and JC-2026 Models

	A	B	n	$\dot{\varepsilon}_0$	α	m	$\dot{\varepsilon}_{t0}$	D	δ	α'	C_0''	C_0'
JC-2019	503.72 MPa	951.63 MPa	0.74	0.001 s ⁻¹	0.0215	0.099	727 s ⁻¹	3.55×10 ⁻¹²	0.7236	-	-	-
JC-2026					0.016	0.547	0.01 s ⁻¹	-	-	-0.1585	0.06288	89.91

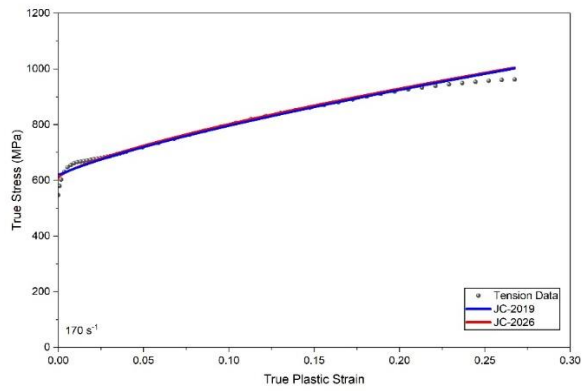
Figure 12 compares the true stress-true plastic strain curves for the 316L stainless steel at various strain rates that were obtained from the tensile experiments and the modified JC models (JC-2019 and JC-2026). As shown in Fig. 12, both JC-2019 and JC-2026 predict the experimental results very well. At low strain rates, i.e., Fig. 12 (a)-(c), the JC-2019 exhibits a better prediction than the JC-2026 model. This difference is possibly attributable to the incorporation of adiabatic heating effects determined by the Boltzmann function, particularly the value of $\dot{\varepsilon}_{t0}$ (727 s⁻¹ in JC-2019 versus 0.01 s⁻¹ in JC-2026 as presented in Table 2), used in both models. However, at higher strain rates, the JC-2026 model slightly outperforms the JC-2019 model, due to the more precise description of adiabatic heating induced thermosoftening effect.



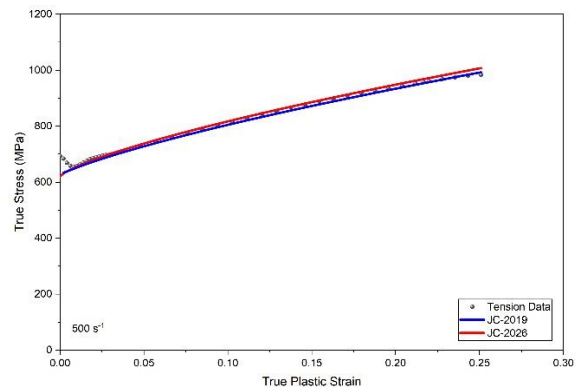
(a) 0.001 s⁻¹



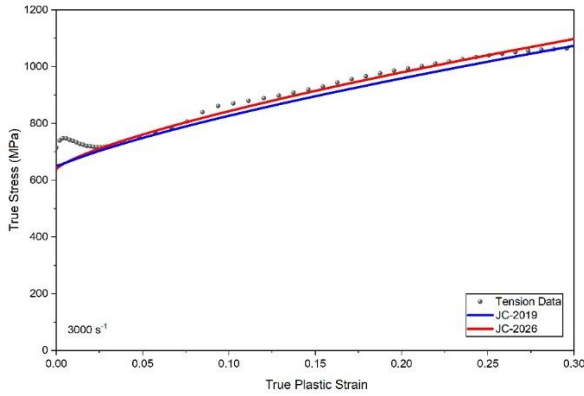
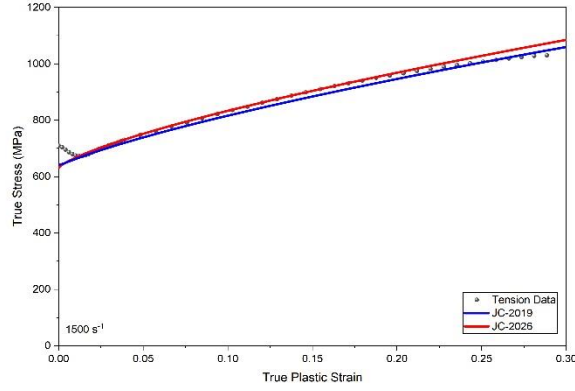
(b) 0.1 s⁻¹



(c) 170 s⁻¹



(d) 500 s⁻¹



(e) 1500 s⁻¹

(f) 3000 s⁻¹

Figure 12. Comparison of tension data and model prediction (JC-2019 and JC -2026) at various strain rates

5 Strain-rate and Strain Dependent TQ Coefficient and Specific Stored Energy

With the JC-2026 model (Equation (9)) and the constants presented in Table 2, both intrinsic and phenomenological TQ coefficients, in integral and differential forms, can be calculated [38],

$$\beta_{int} = \frac{\rho C_v C'_0}{A + \frac{B}{n+1} \varepsilon^n} \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^{\alpha' - \alpha} \quad (10)$$

$$\beta_{diff} = \frac{\rho C_v C'_0}{A + B \varepsilon^n} \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^{\alpha' - \alpha} \quad (11)$$

$$\beta'_{int} = \frac{\rho C_v C'_0}{A + \frac{B}{n+1} \varepsilon^n} \left(1 - \frac{\dot{\varepsilon}_{t0}}{\dot{\varepsilon} + \dot{\varepsilon}_{t0}} \right) \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^{\alpha' - \alpha} \quad (12)$$

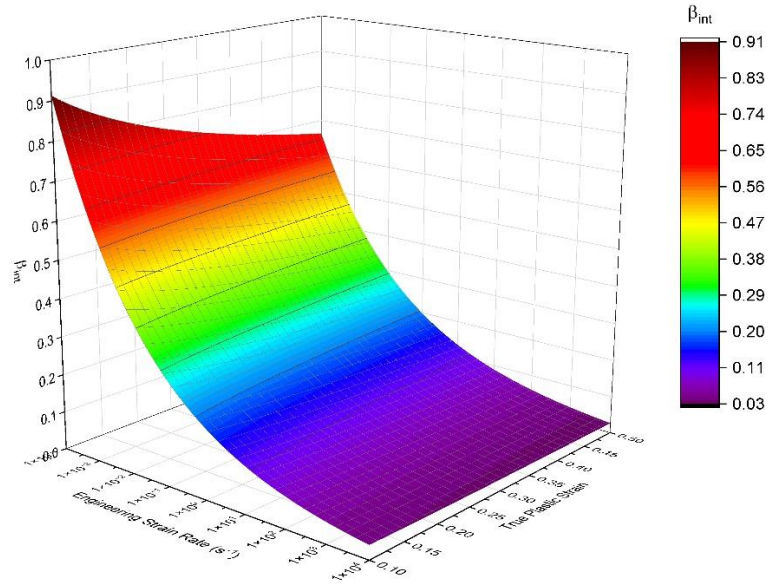
$$\beta'_{diff} = \frac{\rho C_v C'_0}{A+B\epsilon^n} \left(1 - \frac{\dot{\epsilon}_{t0}}{\dot{\epsilon} + \dot{\epsilon}_{t0}}\right) \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0}\right)^{\alpha' - \alpha} \quad (13)$$

where β_{int} and β_{diff} are intrinsic integral and differential TQ coefficients, respectively; β'_{int} and β'_{diff} are phenomenological integral and differential TQ coefficients, respectively; ρ and C_v are density and specific heat of the specimen material, respectively. For 316L stainless steel, $\rho = 8000 \text{ kg/m}^3$, $C_v = 500 \text{ J/kg}^\circ\text{C}$. C'_0 is a constant that can be calculated with

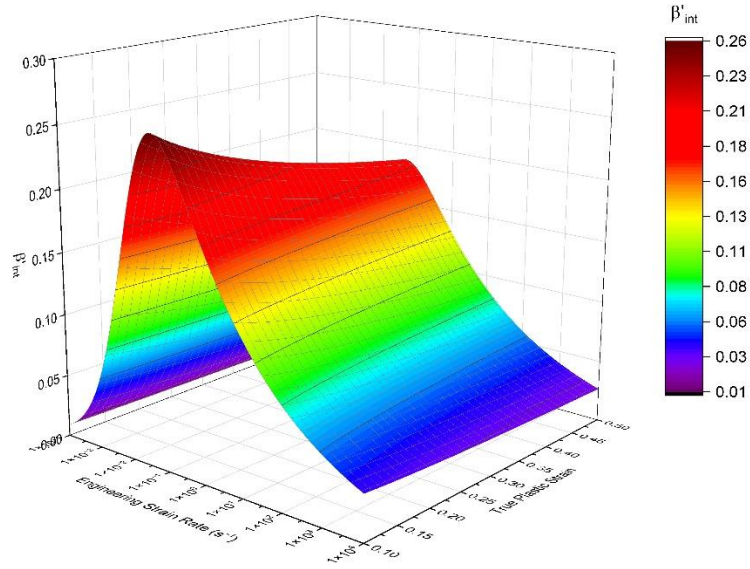
$$C'_0 = C''_0 (T_m - T_0) \quad (14)$$

The value of C'_0 is also listed in Table 2.

Figure 13 shows the strain-rate and strain dependent intrinsic and phenomenological integral TQ coefficients for the 316L stainless. The intrinsic and phenomenological differential TQ coefficients are shown in Fig. 14. The intrinsic TQ coefficients, in either integral or differential form, decrease with increasing strain rates, which is opposite to the observations for 304L-VAR stainless steel [38] [39]. At the same strain rate, the intrinsic TQ coefficients decrease with increasing true plastic strain, which is consistent with 304L-VAR stainless steel. The strain-rate effect on the intrinsic TQ coefficient makes the phenomenological TQ coefficients like a saddle shape, as shown in Figs. 13(b) and 14(b), where the peak value occurs at $\sim 10^{-1} \text{ s}^{-1}$.

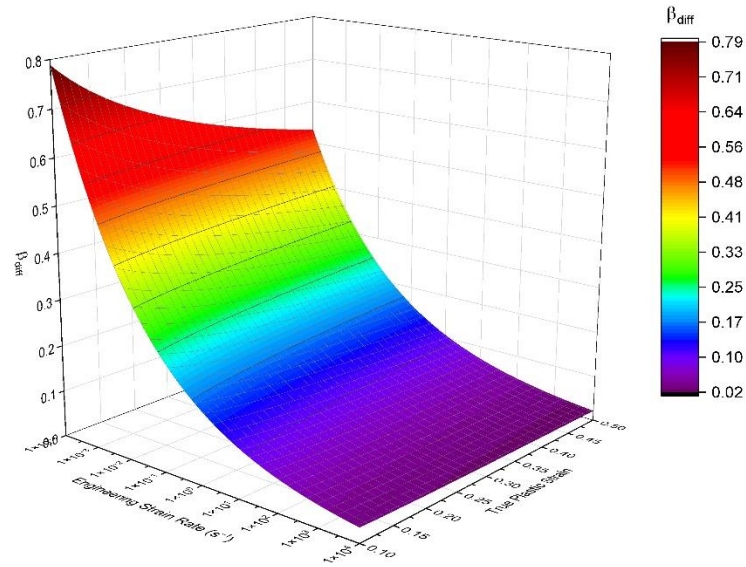


(a) Intrinsic

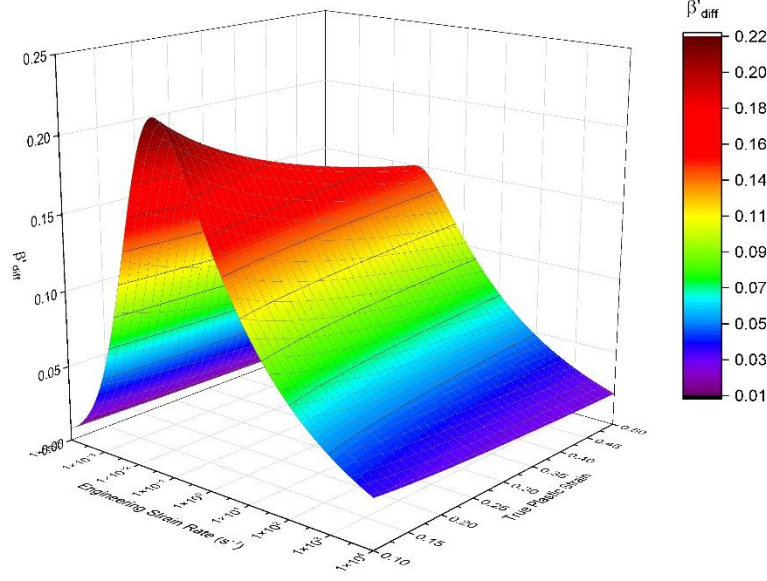


(b) Phenomenological

Figure 13. Integral TQ coefficients for the 316L stainless steel. (a) intrinsic; (b) phenomenological.



(a) Intrinsic;



(b) phenomenological

Figure 14. Differential TQ coefficients for the 316L stainless steel. (a) intrinsic; (b) phenomenological.

The specific stored energy that is normalized by mass can also be calculated as [38],

$$E_s = \varepsilon \left[\frac{1}{\rho} \left(A + \frac{B}{n+1} \varepsilon^n \right) \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^\alpha - C_v C'_0 \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^{\alpha'} \right] \quad (15)$$

The strain-rate and strain dependent specific stored energy for the 316L stainless steel is shown in Fig. 15. The specific stored energy increases with increasing strain rate and true plastic strain. This behavior means that, with increasing strain rate and plastic strain, more energy contributes to microstructural change including phase transformation from austenite to martensite, dislocation, point defects, stacking faults, twins, etc., which may reduce the heat generation during deformation.

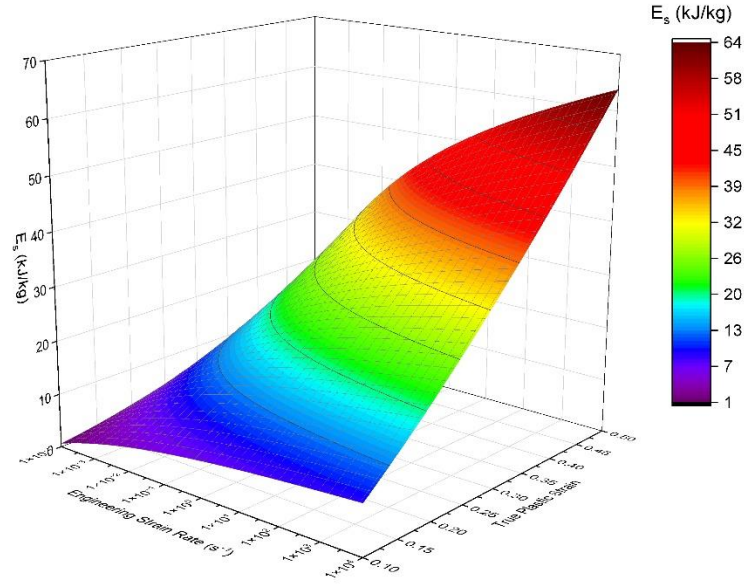


Figure 15. Strain-rate and strain dependent specific stored energy for the 316L stainless steel.

6 Conclusions

In this study, the mechanical response of a 316L stainless steel was characterized in both compression and tension at room temperature and at various strain rates between 10^{-3} s^{-1} and 3000 s^{-1} using an Instron Testing Machine, Drop-Hopkinson (for tension only) bar, and conventional Kolsky bars. The experimental results showed that the tension and compression response exhibited positive strain rate sensitivity. The true failure strain in tension was determined by measuring the minimum diameter of the post-test specimen. In general, the 316L stainless steel is very ductile, and the true tensile failure strain decreased with increasing strain rate. The 316L stainless steel also shows reasonable compression/tension symmetry, particularly at small strains (less than 0.1). The tensile experimental results for the 316 stainless steel were used to fit the modified JC models, JC-2019 and JC-2026, respectively. Both models predicted the experimental results reasonably well although the JC-2026 model accounted for both strain-rate and strain dependent TQ coefficients with a simpler form. Based on the JC-2026 model, the intrinsic and phenomenological strain-rate and strain dependent TQ coefficients for the 316L stainless steel were inversely calculated. For the 316L stainless steel, the intrinsic TQ coefficient in either integral or differential form decreases with increasing strain rates. At the same strain rates, the intrinsic TQ coefficients increase with increasing true plastic strain. The intrinsic TQ was also used to calculate the specific stored energy for the 316L stainless steel. The specific stored energy increases with increasing

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4 strain rate and plastic strain, meaning more energy contributes to microstructural change rather
5 than generating heat.
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22 **Declarations**

23
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30 **References**

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